Introduction to Computational Biology

Course notes: five units, thirty sessions

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Unit 1

Cell Basics and Computing Analogies

The cell is the smallest working unit of life. Here we compare it to things you already know from computing.

S1 Cell theory and prokaryotic vs eukaryotic organisms

Cell theory says three things: all living things are made of cells, the cell is the basic unit of life, and every cell comes from an existing cell. Prokaryotes (bacteria) are small, have no nucleus, and keep their DNA floating in a region called the nucleoid. Eukaryotes (plants, animals, fungi) are larger and have a membrane-bound nucleus plus many organelles.

In depth

- Cell theory was proposed by Schleiden and Schwann (1838-39) and extended by Virchow (1855): every cell arises from a pre-existing cell.
- Prokaryotes: 1-10 micrometre, circular DNA in the nucleoid, 70S ribosomes, peptidoglycan cell wall, divide by binary fission. Eukaryotes: 10-100 micrometre, linear chromosomes inside a nucleus, 80S ribosomes, divide by mitosis.
- Exceptions to remember: viruses are acellular, skeletal muscle cells are multinucleate, and mature red blood cells have no nucleus.
- Endosymbiosis: mitochondria and chloroplasts descend from engulfed bacteria, which is why they have their own circular DNA and 70S ribosomes.

Remember: Prokaryote = open-plan room; eukaryote = building with separate labelled rooms.

S2 Organelles as CPU components analogy

Organelles are tiny compartments, each with a job. Nucleus = CPU control unit (holds the instructions, DNA). Mitochondria = power supply (make ATP, the energy currency). Ribosomes = assembly workers that build proteins. Endoplasmic reticulum and Golgi = data bus and packaging/shipping. Lysosomes = garbage collector. Cell membrane = firewall that decides what enters.

In depth

- Nucleus: double envelope with pores, chromatin (DNA plus histones) and a nucleolus that makes rRNA. Mitochondria: double membrane with folded cristae; glycolysis, Krebs cycle and oxidative phosphorylation yield about 30-32 ATP per glucose.
- Rough ER carries ribosomes and makes secreted proteins; smooth ER makes lipids and detoxifies. Golgi modifies, sorts and packs proteins into vesicles.
- Lysosomes hold digestive enzymes at acidic pH; chloroplasts do photosynthesis; vacuoles store; the cytoskeleton gives shape and tracks for transport.
- Where the analogy breaks: cells have no central clock, work chemically and in parallel, and can repair and copy themselves.

Remember: Analogies help memory, but cells work in parallel and chemically, not by clock cycles.

S3 DNA replication compared to code duplication

Before a cell divides it copies its DNA. Helicase unzips the double helix, primer-laying enzymes start the job, and DNA polymerase adds matching bases (A with T, G with C) to each strand. Each new molecule keeps one old strand and one new strand: this is called semi-conservative replication. Like copying a file, it needs an original, a copier and error checking; polymerase proofreads, like a checksum.

In depth

- Steps: origin of replication opens, helicase unwinds, single-strand binding proteins keep strands apart, topoisomerase relieves supercoils, primase lays an RNA primer, DNA polymerase III extends 5' to 3'.
- Leading strand is made continuously; lagging strand is made as Okazaki fragments, primers are removed by polymerase I and gaps sealed by DNA ligase.
- Accuracy: polymerase proofreads with 3' to 5' exonuclease activity and mismatch repair fixes leftovers; final error rate is about 1 in 10^9 bases.
- Eukaryotes use many origins at once to finish fast; telomerase protects chromosome ends. Computing link: copy with verification, many forks in parallel.

Remember: Error rate is about 1 in a billion bases after proofreading and repair.

S4 Cell division as a concept of parallel computing

Mitosis makes two identical cells (growth, repair); meiosis makes four cells with half the chromosomes (gametes). Mitosis phases: prophase, metaphase, anaphase, telophase, then cytokinesis. One task (a cell) is split into two identical workers that run side by side, like forking a process; billions of cells dividing at once is massive parallelism.

In depth

- Cell cycle: G1 (growth), S (DNA synthesis), G2 (preparation), M (mitosis). Checkpoints at G1/S, G2/M and the spindle checkpoint are controlled by cyclins and CDKs.
- Mitosis: prophase (chromosomes condense), metaphase (line up at the plate), anaphase (sister chromatids pulled apart), telophase (two nuclei form), then cytokinesis.
- Meiosis has two divisions, with crossing over in prophase I and independent assortment, so gametes are genetically different and haploid (n).
- Computing view: dividing is like fork(); many cells dividing is a multi-threaded system. Cancer is like a loop that lost its exit condition.

Remember: Checkpoints in the cell cycle act like assertions: if DNA is damaged, division stops.

S5 Genetic algorithms for biological optimization

A genetic algorithm (GA) borrows evolution to search for good solutions. Steps: create a random population of candidate solutions, score each with a fitness function, select the best, combine them (crossover), randomly tweak some (mutation), and repeat for many generations. Used for protein design, drug molecule search and scheduling problems where trying every option is impossible.

In depth

- Steps: encode each solution as a chromosome (bit string or array), score fitness, select parents (roulette wheel or tournament), crossover (swap parts, e.g. single point), mutate (flip a few genes, about 1% rate), form the next generation, stop after N generations or when fitness is good enough.
- Pseudocode: initialise population; repeat { evaluate; select; crossover; mutate; replace } until stop. Elitism keeps the best individual unchanged.
- Biology example: choose a small set of genes that best separates cancer from normal. Fitness = classification accuracy minus a penalty for using many genes.
- Pros: needs no gradient, handles discrete and messy search spaces. Cons: slow, can converge early to a poor solution, and needs tuning of population size and rates.

Remember: Population, fitness, selection, crossover, mutation: memorise these five words.

S6 Evolutionary computing in cell/tissue modelling

Evolutionary computing is the wider family that includes GAs, evolution strategies and genetic programming. In biology it is used to fit model parameters (for example, how fast a cell grows), to design gene networks and to simulate how tissues form from many interacting cells. Natural evolution itself is the original algorithm that these methods imitate.

In depth

- Evolutionary computing family: genetic algorithms, genetic programming (evolves programs), evolution strategies, differential evolution.
- Parameter fitting: evolve the rate constants of a differential equation model until the simulation matches lab data; fitness = error between simulated and measured values.
- Tissue modelling: agent-based models treat each cell as an agent with rules (divide, move, die, signal). Evolution can tune those rules to reproduce tumour growth or wound healing.
- Limits: heavy computing, and a model that fits the data may still be biologically wrong, so it needs experimental validation.

Remember: Good when the search space is huge and no neat formula exists.

Unit 2

Biomolecules as Data Structures

Life is built from four main molecule classes. We treat them as storage, structure and information formats.

S1 Carbohydrates and lipids as energy and membrane storage

Carbohydrates (sugars, starch, glycogen, cellulose) are quick energy and structure; glucose is the cell's main fuel. Lipids (fats, oils, phospholipids, steroids) store more energy per gram and form the two-layer membrane (phospholipid bilayer) with water-loving heads outside and water-fearing tails inside. Think of carbs as RAM (fast, short-term) and fats as hard disk (dense, long-term).

In depth

- Carbohydrates: monosaccharides (glucose, fructose), disaccharides (sucrose, maltose, lactose) joined by glycosidic bonds, polysaccharides (starch in plants, glycogen in animals, cellulose in cell walls). Humans cannot digest cellulose (beta-1,4 links).
- Lipids: triglycerides (glycerol plus 3 fatty acids), saturated (solid, no double bonds) vs unsaturated (liquid, double bonds), phospholipids, steroids such as cholesterol.
- Membrane: fluid mosaic model. Phospholipid bilayer with heads out, tails in; cholesterol controls fluidity; proteins transport and signal; glycoproteins help cell recognition (e.g. blood groups).
- Energy: complete oxidation gives about 4 kcal/g for carbohydrate and 9 kcal/g for fat, so fat is the compact long-term store.

Remember: Fats give about 9 kcal/g, carbs about 4 kcal/g.

S2 Proteins and nucleic acids as functional data structures

Proteins are chains of amino acids that fold into working machines: enzymes, transporters, antibodies, structural fibres. Nucleic acids (DNA, RNA) store and pass on instructions. In computing terms: DNA is the read-only master archive, RNA is a working copy, a protein is the executed program. A protein sequence is like a string; its 3D fold is the object it becomes.

In depth

- Amino acid: central carbon with amino group, carboxyl group, hydrogen and a side chain R. The 20 types are grouped as non-polar, polar, acidic and basic by their R group.
- Protein roles: enzymes, transport (haemoglobin), defence (antibodies), structure (collagen, keratin), hormones (insulin), movement (actin, myosin).
- Nucleic acids: nucleotide = sugar + phosphate + base. DNA has deoxyribose and A, T, G, C; RNA has ribose and A, U, G, C. ATP is also a nucleotide.
- Data view: a protein is a string over 20 letters, DNA is a string over 4 letters (2 bits per base, so the human genome is about 750 MB raw). Stored in FASTA format.

Remember: Central dogma: DNA, then RNA, then protein.

S3 Enzymes as biological catalysts

Enzymes are mostly proteins that speed up reactions without being used up. The substrate fits into the active site (lock and key, or the more accurate induced-fit idea) and the activation energy drops. Speed depends on temperature, pH and substrate amount; too hot and the enzyme denatures. Inhibitors can block them, which is how many drugs work.

In depth

- Classes (EC numbers): oxidoreductases, transferases, hydrolases, lyases, isomerases, ligases.
- Michaelis-Menten: v = Vmax[S] / (Km + [S]). Low Km means high affinity for the substrate; Vmax is reached when all enzyme is saturated.
- Inhibition: competitive (binds active site, Km rises, Vmax unchanged), non-competitive (binds elsewhere, Vmax falls). Allosteric regulation and feedback inhibition control pathways.
- Cofactors (metal ions) and coenzymes (vitamin derivatives like NAD) help. Optimum conditions: about 37 C, pH differs (pepsin 2, trypsin 8). Examples: amylase, catalase, DNA polymerase; ACE inhibitors are enzyme-blocking drugs.

Remember: Like a function: takes an input (substrate), returns an output (product), reusable.

S4 Nucleic acid structure and function relationships

DNA is a double helix: two antiparallel strands of nucleotides (sugar, phosphate, base) held by hydrogen bonds, A-T with two bonds and G-C with three. RNA is usually single-stranded, uses ribose and uracil in place of thymine. Types: mRNA carries the message, tRNA brings amino acids, rRNA builds ribosomes. Structure decides function: the stable helix stores data, the flexible single strand works.

In depth

- Watson and Crick (1953) built the model using Franklin's X-ray data. Chargaff's rules: A equals T and G equals C.
- B-DNA: right-handed, 10 base pairs per turn, 3.4 angstrom rise per pair, 34 angstrom pitch, 20 angstrom diameter, with major and minor grooves where proteins read the bases. Strands are antiparallel (5' to 3' opposite 3' to 5').
- RNA: mRNA carries the gene copy (eukaryotes add a 5' cap, poly-A tail and splice out introns), tRNA has a cloverleaf shape with an anticodon, rRNA is the ribosome core. Some RNAs act as enzymes (ribozymes).
- Melting temperature (Tm) rises with GC content because G-C has 3 hydrogen bonds against 2 for A-T.

Remember: Base pairing is what makes copying and reading possible.

S5 BLAST tool for sequence database queries

BLAST (Basic Local Alignment Search Tool) takes your DNA or protein sequence and finds similar ones in huge databases such as NCBI's. It breaks the query into short words, finds matches, extends them into alignments and scores them. Look at the E-value (lower means more significant), percent identity and query coverage. It is how you guess what a new gene does: find its known relatives.

In depth

- How it works: split the query into short words, find matching words in the database, extend them into local alignments, score with a matrix (BLOSUM62 for protein), report the best hits.
- Programs: blastn (DNA vs DNA), blastp (protein vs protein), blastx (translated DNA vs protein), tblastn, tblastx.
- Reading the result: E-value is the number of hits expected by chance (below 1e-5 usually significant), bit score is size-independent, percent identity and query coverage show how similar and how much matched.
- Uses: identify an unknown gene or species, find homologs to guess function, check primers. Limits: heuristic (can miss weak matches) and results depend on the database.

Remember: It is basically Google search for sequences.

S6 Human Genome Project as big data genomics

The Human Genome Project (1990-2003) read all about 3 billion base pairs of human DNA, revealing roughly 20,000 protein-coding genes. It proved that biology needs data storage, assembly algorithms and databases. It started modern genomics, cheap sequencing and personalised medicine, and raised privacy and ethics questions about genetic data.

In depth

- Timeline: launched 1990 (US DOE and NIH with UK, Japan, France, Germany, China), draft in 2001, declared complete in 2003; the remaining gaps were closed by the telomere-to-telomere project in 2022.
- Methods: hierarchical shotgun (public project) and whole-genome shotgun (Celera). Software had to assemble millions of short reads, a huge computing problem.
- Findings: about 3 billion base pairs, roughly 20,000-25,000 genes, only about 1.5 percent codes for protein, and any two people are about 99.9 percent identical.
- Impact: data stored in GenBank, Ensembl and UCSC browser; led to next-generation sequencing, personalised medicine and falling cost (billions then, under $1000 now). About 5 percent of the budget went to ethical, legal and social issues.

Remember: Raw data in; assembly, annotation and analysis software out.

Unit 3

Protein Structures and Prediction Algorithms

How a sequence becomes a shape, and how computers predict that shape.

S1 Protein synthesis and primary structure

Transcription copies a gene from DNA into mRNA in the nucleus. Translation happens at the ribosome: each three-base codon on mRNA is matched by tRNA to one amino acid, building a chain until a stop codon. The order of amino acids joined by peptide bonds is the primary structure, and it alone determines how the protein folds.

In depth

- Transcription: RNA polymerase binds the promoter (TATA box in eukaryotes), builds mRNA 5' to 3' on the template strand, stops at a terminator. Eukaryotic mRNA is processed: 5' cap, poly-A tail, introns spliced out.
- Translation: initiation (small subunit finds AUG, codes methionine), elongation (tRNA enters the A site, peptide bond forms, ribosome shifts through P and E sites), termination (UAA, UAG or UGA, release factors).
- Genetic code: triplet, degenerate (64 codons for 20 amino acids), universal, non-overlapping, no punctuation.
- Primary structure is read N-terminus to C-terminus and decides the fold (Anfinsen). Mutations: silent, missense (sickle cell: Glu to Val), nonsense, frameshift.

Remember: 20 amino acids, 64 codons, so the code is redundant.

S2 Secondary, tertiary and quaternary organisation

Secondary: local patterns held by hydrogen bonds, mainly alpha helices and beta sheets. Tertiary: the full 3D fold of one chain, stabilised by hydrophobic packing, ionic bonds, disulfide bridges. Quaternary: several chains joined into one complex, like haemoglobin with four subunits. Each level builds on the one below.

In depth

- Secondary: alpha helix (3.6 residues per turn, hydrogen bond from residue i to i+4) and beta sheet (parallel or antiparallel strands), plus turns and loops. A Ramachandran plot shows allowed phi and psi angles.
- Tertiary: hydrophobic core, hydrogen bonds, ionic bonds, disulfide bridges, van der Waals forces. Chains may have domains. Chaperones help folding.
- Quaternary: subunits assemble, e.g. haemoglobin (two alpha, two beta) and antibodies.
- Types: fibrous (collagen, keratin, long and strong) vs globular (enzymes, compact). Misfolding causes prion disease and is linked to Alzheimer's.

Remember: Primary = text, secondary = formatting, tertiary = whole document, quaternary = bound book.

S3 PyMOL for 3D protein visualisation

PyMOL is software to view and render molecules from PDB files (the Protein Data Bank stores solved structures, each with an ID like 1TUP). You can show cartoon, stick, surface views, colour by chain or residue, measure distances and make images. Basic commands: fetch 1tup, show cartoon, color red, ss h, zoom.

In depth

- The PDB database stores solved structures; each entry has an ID (e.g. 1TUP). A PDB file lists ATOM lines with atom name, residue, chain and x, y, z coordinates.
- Common commands: fetch 1tup, show cartoon or sticks or surface, color by chain, select, label, distance, zoom, png to save an image.
- Uses: locate the active site, measure hydrogen bonds (about 3 angstrom), superimpose two structures with align, view ligand binding, mutate a residue.
- Alternatives: UCSF Chimera, Mol* in the browser. Free educational versions of PyMOL exist.

Remember: Seeing the active site helps understand how a drug or ligand binds.

S4 Secondary structure prediction methods

Goal: predict helix, sheet or coil for each residue from sequence alone. Early methods: Chou-Fasman (amino acid propensities) and GOR (statistics of neighbours). Later: neural networks (PSIPRED) using evolutionary profiles, reaching about 80 percent accuracy. Modern deep learning tools such as AlphaFold go further and predict the full 3D structure.

In depth

- Generations: Chou-Fasman (residue propensities, 50-60 percent), GOR (looks at a window of neighbours), PSIPRED (neural network on evolutionary profiles, about 80 percent).
- Input is the sequence (or multiple alignment); output is H, E or C for each residue. Accuracy is reported as Q3.
- Tertiary prediction: homology modelling (SWISS-MODEL), threading, ab initio, and AlphaFold2 (CASP14, 2020), which predicts near-experimental structures. The 2024 Chemistry Nobel went to Baker, Hassabis and Jumper for protein design and prediction.

Remember: Input is a string of letters, output is a string of H, E, C labels.

S5 Python programming for structure analysis

Python with Biopython lets you parse PDB files, loop over chains and residues, read atom coordinates, count residues, and compute distances between atoms. Typical steps: parse the file with PDBParser, iterate model, chain, residue, atom, then use NumPy for calculations and matplotlib to plot, for instance a Ramachandran-style or distance plot.

In depth

- Biopython workflow: install with pip, use PDBParser to get_structure from a file, then loop structure, model, chain, residue, atom.
- Useful calls: atom.get_coord() for x, y, z; subtracting two atoms gives their distance; PPBuilder extracts the sequence; DSSP assigns secondary structure.
- Analysis ideas: count residue types, find atoms within 4 angstrom of a ligand, compute radius of gyration, inspect B-factors (flexibility), plot with matplotlib and NumPy.
- Structure is just nested data: model, chain, residue, atom with coordinates.

Remember: Structure is just a nested list of atoms with x, y, z numbers.

S6 Basics for drug design

Drug design finds molecules that bind a target protein and change its action. Structure-based design uses the 3D target and docking software to test how a ligand fits; ligand-based design uses known active molecules. Key ideas: binding site, affinity, selectivity, then lead optimisation and ADMET (absorption, distribution, metabolism, excretion, toxicity) checks, followed by clinical trials.

In depth

- Pipeline: target identification and validation, hit discovery (high-throughput or virtual screening), lead optimisation, preclinical tests, Phase I-III trials, approval. It takes 10-15 years and over a billion dollars.
- Structure-based: dock ligands into the binding pocket (AutoDock Vina) and rank by score. Ligand-based: pharmacophore and QSAR from known actives.
- Lipinski rule of five for oral drugs: mass under 500, logP under 5, at most 5 H-bond donors, at most 10 acceptors.
- Success stories: captopril (ACE), HIV protease inhibitors, imatinib. ADMET filters remove toxic or poorly absorbed molecules early.

Remember: Docking = trying keys in a lock and scoring the fit.

Unit 4

AI/ML Foundations in Biology

Learn the machine learning basics needed to work with biological data.

S1 Overview of AI for biological big data

Biology now produces huge data: genomes, proteomes, images, patient records. AI means machines doing tasks that need intelligence; machine learning (ML) is the part that learns patterns from data; deep learning is ML using many-layered neural networks. Uses include gene finding, disease prediction, protein folding and image analysis.

In depth

- AI = machines doing tasks needing intelligence; ML = learning patterns from data; deep learning = ML with many-layered neural networks.
- Biological big data has volume, variety (sequences, structures, images, health records) and velocity (fast sequencers). GenBank and other databases keep growing rapidly.
- Tasks: classification, prediction, clustering, generation. Examples: DeepVariant for variant calling, AlphaFold for structure.
- Why now: more data, powerful GPUs, better algorithms.

Remember: AI contains ML, which contains deep learning.

S2 Supervised and unsupervised learning basics

Supervised learning trains on labelled examples (tumour or normal) to predict labels: classification (categories) and regression (numbers). Unsupervised learning finds structure in unlabelled data: clustering groups similar genes or patients, and PCA reduces dimensions. Models are split into training and test sets to check they generalise; overfitting means memorising instead of learning.

In depth

- Supervised algorithms: linear and logistic regression, decision trees, random forest, SVM, k-nearest neighbours, neural networks. Classification gives a category; regression gives a number.
- Unsupervised: k-means and hierarchical clustering, PCA, t-SNE or UMAP. Example: clustering expression data to discover cancer subtypes.
- Data split: training (fit), validation (tune), test (final check), e.g. 70/15/15. Overfitting = great on training, poor on new data; underfitting = too simple. Regularisation reduces overfitting.
- Also exist: semi-supervised and reinforcement learning.

Remember: Labels given = supervised. No labels = unsupervised.

S3 Feature representations for sequences and omics

Models need numbers. DNA can be one-hot encoded (A=1000, C=0100...), split into k-mers (short substrings) or embedded by learned vectors. Omics data (genomics, transcriptomics, proteomics, metabolomics) become a matrix: rows are samples, columns are features like gene expression. Good features matter more than the algorithm.

In depth

- One-hot encoding turns A, C, G, T into vectors like 1000. k-mer counts treat sequences as words (k=3 gives 64 features). Learned embeddings and protein language models (e.g. ESM) give richer vectors.
- Omics: genomics (DNA), transcriptomics (RNA), proteomics, metabolomics. Data forms a matrix with samples as rows and genes as columns.
- Curse of dimensionality: far more features than samples. Fix with feature selection (variance, LASSO) or PCA.
- Normalise first (e.g. TPM, log transform) so large values do not dominate.

Remember: Normalise and select features first, to avoid 20,000 columns for 50 samples.

S4 AI data pipeline: preprocess, train, evaluate

Pipeline: collect data, clean it (missing values, outliers, batch effects), preprocess (scale, encode), split train/validation/test, train a model, tune hyperparameters, evaluate. Metrics: accuracy, precision, recall, F1, ROC-AUC. Use cross-validation for small datasets, and never let test data leak into training.

In depth

- Steps: collect, clean (missing values, outliers, batch effects), preprocess (scale, encode), split, train, tune hyperparameters (grid search), evaluate, deploy and monitor.
- Confusion matrix: TP, FP, TN, FN. Precision = TP/(TP+FP); recall (sensitivity) = TP/(TP+FN); F1 is their harmonic mean; ROC-AUC measures ranking quality.
- Cross-validation (k-fold) is used for small datasets. Imbalanced data needs class weights or SMOTE. Data leakage (test info reaching training) gives falsely good results.
- Tools: Python with pandas and scikit-learn.

Remember: Precision = how many predicted positives are right; recall = how many real positives are found.

S5 Applications in genomics and systems biology

In genomics, ML predicts gene variants' effects, finds promoters and classifies cancer types. In systems biology, it models gene regulatory networks, metabolic pathways and how many components interact; it helps predict cell behaviour after a drug or mutation. Network and graph methods represent genes and proteins as nodes with links.

In depth

- Genomics uses: variant effect prediction, gene and promoter finding, GWAS, cancer type classification, CRISPR guide design.
- Systems biology studies the whole network: gene regulatory networks, signalling and metabolic pathways (flux balance analysis). Genes and proteins are nodes, interactions are edges, hubs are highly connected nodes.
- Combine ordinary differential equation models with ML; cluster single-cell RNA-seq data into cell types; predict drug response of a cell line.
- Goal: a computational 'digital twin' of a cell to test interventions.

Remember: Systems biology looks at the whole network, not one gene.

S6 Benefits and limitations of AI in biology

Benefits: speed, finding patterns humans miss, cheaper screening, scalable analysis. Limitations: needs large, good-quality labelled data; bias if data lack diversity; black-box models are hard to explain; results may not generalise; lab validation is still required. AI supports scientists, it doesn't replace experiments.

In depth

- Benefits: speed, scalability, finding hidden patterns, cheaper screening, earlier diagnosis.
- Limits: garbage in, garbage out; small samples with many features; batch effects; biased data; black-box models; irreproducible results; heavy compute and energy use.
- Mitigation: diverse data, external validation on new cohorts, explainable AI (SHAP, saliency maps), keeping a human in the loop, and confirming findings by experiment.
- Key idea: a model finds correlation, biology needs causation.

Remember: Correlation found by a model is not proof of a biological cause.

Unit 5

Deep Learning, Healthcare and Ethics

Neural networks in medicine, and the responsibilities that come with them.

S1 Deep learning for images, signals and sequences

A neural network has layers of connected nodes with weights learned by backpropagation. CNNs (convolutional) find patterns in images such as X-rays and microscope slides. RNNs, LSTMs and transformers handle sequences like DNA, proteins and text. ECG and EEG signals are time-series, handled by 1D CNNs or sequence models.

In depth

- Neuron: weighted sum of inputs plus bias passed through an activation (ReLU, sigmoid, softmax). Layers stack neurons; a loss function measures error; gradient descent with backpropagation updates weights over many epochs.
- CNN: convolution filters detect edges and patterns, pooling shrinks the image, fully connected layers classify. Used for X-rays, slides, microscopy. A 1D CNN can find DNA motifs.
- RNN and LSTM process sequences step by step; transformers use attention to relate all positions and power protein language models. ECG and EEG are time-series signals.
- Also: transfer learning (reuse a trained model), autoencoders, GANs. Needs large data and GPUs.

Remember: Images: CNN. Sequences: RNN or transformer.

S2 AI in drug discovery and biotechnology

AI predicts protein structure (AlphaFold), screens millions of compounds virtually, designs new molecules (generative models), predicts toxicity, and suggests drug repurposing. In biotech it helps design enzymes, optimise fermentation and engineer crops. This cuts the years and cost of finding a drug candidate.

In depth

- Drug discovery stages helped by AI: target finding, virtual screening of millions of molecules, generative design (VAEs, GANs, diffusion), ADMET prediction, repurposing existing drugs.
- AlphaFold database provides predicted structures for over 200 million proteins, speeding up target analysis. AI-designed candidates have entered clinical trials.
- Biotechnology: enzyme engineering, CRISPR guide prediction, synthetic biology design, fermentation optimisation, crop improvement.
- Value: cuts cost and years from early discovery, but lab and clinical validation are still essential.

Remember: Shortlist with AI, confirm in the lab.

S3 AI-assisted medical diagnostics and imaging

Deep learning reads chest X-rays, CT, MRI, retina scans and pathology slides to detect pneumonia, tumours, diabetic retinopathy and more, often matching specialists on narrow tasks. It acts as a second opinion, flags urgent cases and reduces workload, while the doctor makes the final decision.

In depth

- Data: X-ray, CT, MRI, ultrasound, retinal fundus images, histopathology slides, dermoscopy. Tasks: classification, detection, segmentation (U-Net), report generation.
- Example: automated diabetic retinopathy screening approved by the US FDA in 2018.
- Metrics: sensitivity (finds disease), specificity (avoids false alarms). Heat maps like Grad-CAM show where the model looked.
- Issues: models fail on data from a different hospital (dataset shift), need regulatory approval (FDA, CDSCO), and must stay a support tool. Helpful where specialists are scarce, such as rural areas.

Remember: Tools are 'decision support', not a doctor replacement.

S4 AI for integrated bioinformatics and health data

Patient care generates electronic health records, genomic data, lab values, wearables and images. Integrating these (multi-omics plus clinical data) allows precision medicine: predicting disease risk, choosing a therapy for a patient's genetic profile. Challenges are different formats, missing data and privacy.

In depth

- Sources: electronic health records, genomes, imaging, lab tests, wearables. Standards help integration: FHIR and HL7 for records, DICOM for images.
- Precision medicine: pick drug and dose using the patient's genes, e.g. CYP2C19 variants affect clopidogrel response. Multi-omics plus clinical data improves risk prediction.
- Techniques: knowledge graphs, and federated learning, where the model trains at each hospital so raw data never leaves.
- Challenges: missing values, different formats, privacy, interoperability. India's Ayushman Bharat Digital Mission is building such a health data system.

Remember: Combine many data types for a fuller patient picture.

S5 Real-time AI for monitoring and decision support

Sensors and wearables stream heart rate, glucose and oxygen. Models detect abnormal patterns live: ICU early-warning for sepsis, arrhythmia alerts, insulin dosing support. Needs low latency, reliability and few false alarms, since too many alerts cause alarm fatigue.

In depth

- Wearables and bedside sensors stream heart rate, oxygen and glucose. Models flag abnormal patterns as they happen, often on the device itself (edge computing).
- Examples: ICU sepsis early warning, smartwatch arrhythmia alerts, closed-loop insulin delivery (artificial pancreas), outbreak surveillance.
- Requirements: low latency, reliability, few false alarms (too many cause alarm fatigue), explanations the clinician can check.
- Decision support presents suggestions; the clinician stays responsible.

Remember: Speed and trust are as important as accuracy.

S6 Ethical and social issues of AI in biology

Key issues: patient privacy and consent, data security, bias that harms under-represented groups, explainability, who is responsible if AI errs, fair access to technology, and misuse (such as dangerous biological design). Principles: fairness, transparency, accountability, human oversight, and following rules like HIPAA, GDPR and India's DPDP Act.

In depth

- Privacy and consent: health and genetic data are sensitive; anonymisation can fail; people must know how data is used. Genetic discrimination (insurance, jobs) is a risk.
- Bias and fairness: training data mostly from European ancestry or certain skin tones gives worse results for others. Accountability: who is responsible when AI errs? Transparency: can the decision be explained?
- Access and misuse: digital divide in who gets AI care; dual-use risk where tools could help design harmful biological agents.
- Rules: HIPAA (US), GDPR (EU), India's DPDP Act 2023, ICMR ethical guidelines for AI in health. Principles: autonomy, beneficence, non-maleficence, justice.

Remember: Ask 'who benefits, who is harmed, who is accountable?'